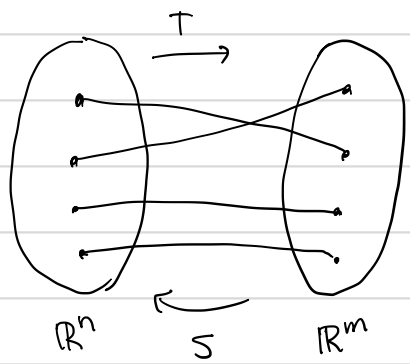
Lecture 14: Inverse transf. Basis + dimension

The <u>inverse</u> of $T: \mathbb{R}^n \to \mathbb{R}^m$:

a transformation $S: \mathbb{R}^m \to \mathbb{R}^n$

satisfying:

1) $S(T(v)) = v$ for all v in $\mathbb{R}^n$
2) $T(S(w)) = w$ for all w in $\mathbb{R}^m$



<u>Write</u>: $S = T^{-1}$.

<u>Facts</u>:

- T^{-1} exists if, and only if,
  1) T is one-to-one, and
  2) T is onto.

- T linear $\Rightarrow$ T^{-1} linear

  $T(v) = A \cdot v$ $\qquad$ $T^{-1}(w) = B w$

  * T^{-1} exists $\Rightarrow$ $m = n$
  * $T^{-1}(T(v)) = B \cdot A v = I \cdot v$
  * $T(T^{-1}(w)) = A \cdot B \cdot w = I w$

$$\boxed{B = A^{-1}}$$

<u>Def</u>: $v_1, v_2, \dots, v_m$ form a <u>basis</u> for a subspace S of $\mathbb{R}^n$ if

1) $v_1, \dots, v_m$ span S
2) $v_1, \dots, v_m$ are independent.

If $v \in S$, then

$$v = a_1 v_1 + \dots + a_m v_m. \qquad (1)$$

If also

$$v = b_1 v_1 + \dots + b_m v_m,$$

then

$$0 = v - v = (a_1 - b_1) v_1 + \dots + (a_m - b_m) v_m.$$

The v_i are indep., so (2)

$$a_1 = b_1, \dots, a_m = b_m.$$

<u>Example</u>: The columns of

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

form a basis for $\mathbb{R}^2$.

<u>Example</u>: The columns of

$$U = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

span col(U), but do not form a basis. $\boxed{\text{The columns with pivots do}}$

<u>Theorem</u>: If $v_1, \dots, v_m$ and $w_1, \dots, w_n$ are bases for a subspace S of $\mathbb{R}^N$, then $m=n$.

<u>Def</u>: The <u>dimension</u> of a subspace is the number of elements in any basis for that subspace.

<u>Proof of theorem</u>: Suppose $n>m$.

$$\underbrace{\begin{bmatrix} | & & | \\ v_1 & \cdots & v_m \\ | & & | \end{bmatrix}}_{V} \underbrace{\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} | & & | \\ w_1 & \cdots & w_n \\ | & & | \end{bmatrix}}_{W}$$

$n>m \Rightarrow Ax=0$ for some $x \neq 0$

$\Rightarrow \underbrace{VAx}_{Wx}=0$

$\Rightarrow w_1, \dots, w_n$ are not independent. ∎